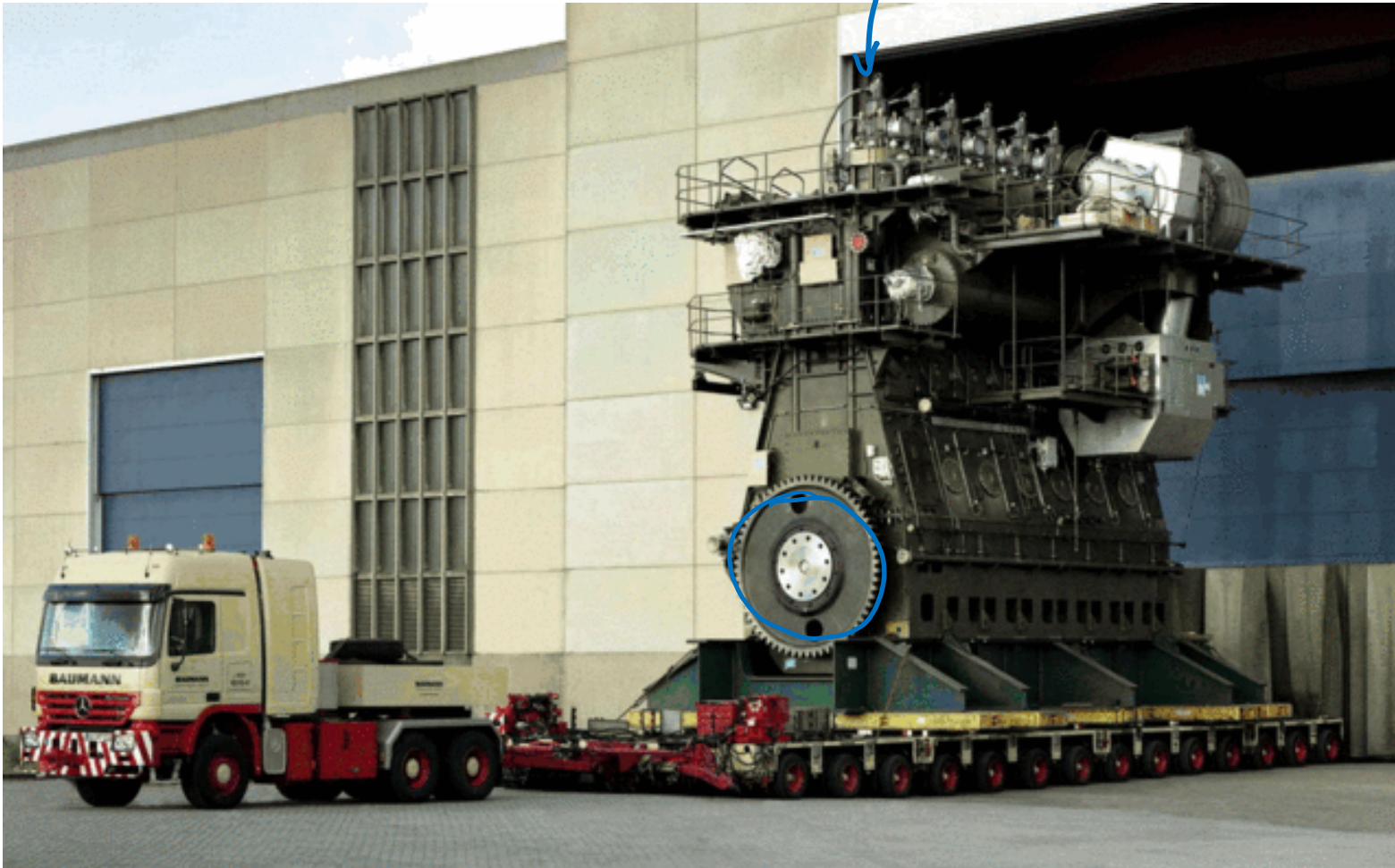




ME 300: Engineering Thermodynamics

Example: Equilibrium

Ship engine



~ 600 RPM

http://telstarlogistics.typepad.com/_a/6a00d834543b6069e20133f4dd923e970b-640wi

Ship engine

$$600 \text{ RPM} = 10 \text{ RPS} \rightarrow t_{\text{cycle}} = \frac{1}{10} \text{ sec}$$

$$t_{\text{stroke}} = \frac{1}{20} \text{ sec}$$

$$L = 1 \text{ meter}$$

$$N_{\text{ave}} = \frac{1}{1/20} = 20 \text{ m/s}$$



Is the quasi-equilibrium assumption reasonable for this compression?

- a) Yes
- b) No

Ship engine

$$P_2 > P_1$$

$$c = 340 \text{ m/s}$$

$$v = 20 \text{ m/s}$$